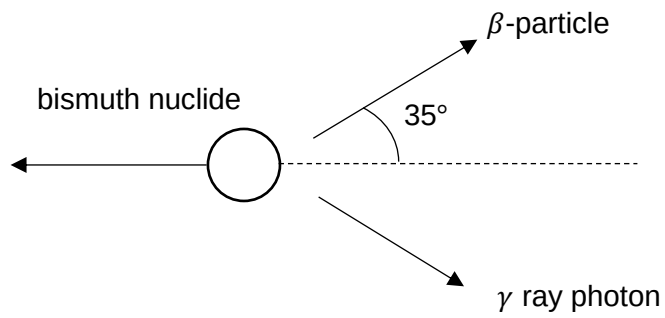


9	Some elements that are normally stable, such as lead (Pb), have isotopes which are radioactive. The nucleus $^{214}_{82}\text{Pb}$ is one such isotope of lead.	
	(a)	State what is meant by isotopes.
		
		
		
		
		 [2]
		Two or more forms of the same element, having the <u>same number of protons</u> but <u>different number of neutrons</u> in their nuclei.
		<i>This was well done. However, students should not include electrons in the definition.</i>
	(b)	A nucleus of $^{214}_{82}\text{Pb}$ decays by β emission into $^{214}_{83}\text{Bi}$. This bismuth nuclide is itself radioactive with an unusual decay pattern. Sometimes it decays by α emission into tellurium (Tl) and sometimes by β emission into polonium (Po). Write the nuclear equations for these two decays of $^{214}_{83}\text{Bi}$.
		α emission:
		β emission: [2]
		$^{214}_{83}\text{Bi} \rightarrow ^{210}_{81}\text{Tl} + ^4_2\text{He}$ $^{214}_{83}\text{Bi} \rightarrow ^{214}_{84}\text{Po} + ^0_{-1}\text{e}$
		<i>Generally well done, although a small number of students were not clear which decay to represent.</i>
	(c)	The two decay patterns of the $^{214}_{83}\text{Bi}$ each give rise to γ ray photons. Suggest why each of these photons have different energies.
		
		
		
		
		 [2]
		The <u>total mass/binding energies of the products</u> of these two decay patterns are <u>different</u> .
		Hence the <u>difference in total mass/binding energies before and after decay is different</u> for these two decay patterns and the energies released in the γ ray photons are different.
		<i>Some phrases are unclear or nonsensical, e.g. the mass defect of a decay, only nuclides can have a mass defect or the responses did not indicate clearly which mass or binding energy they are referring to.</i>
	(d)	A stationary $^{214}_{82}\text{Pb}$ decays by β emission into $^{214}_{83}\text{Bi}$ as shown in Fig. 9.1.

**Fig. 9.1**

The energy of the β -particle is 0.74 MeV and the energy of the bismuth nuclide is 5.5 eV.

(i) Determine the momentum of

1. the β -particle

momentum = kg m s⁻¹ [1]

2. and the bismuth nuclide

momentum = kg m s⁻¹ [1]

Momentum of β -particle = $\sqrt{2mE}$

$$= \sqrt{2 \times 9.11 \times 10^{-31} \times 0.74 \times 10^6 \times 1.60 \times 10^{-19}} = 4.645 \times 10^{-22} \text{ kg m s}^{-1}$$

Momentum of bismuth nuclide = $\sqrt{2mE}$

$$= \sqrt{2 \times 214 \times 1.66 \times 10^{-27} \times 5.5 \times 1.60 \times 10^{-19}} = 7.907 \times 10^{-22} \text{ kg m s}^{-1}$$

Many responses used the wrong masses.

		(ii)	Hence, using the principle of conservation of linear momentum that the wavelength of the γ ray photon is 1.36 pm.	
				[4]
			Using conservation of linear momentum Horizontal component of momentum of γ ray photon = $7.907 \times 10^{-22} - 4.645 \times 10^{-22} \cos 35^\circ$ $= 4.102 \times 10^{-22} \text{ kg m s}^{-1}$ Vertical component of momentum of γ ray photon = $4.645 \times 10^{-22} \sin 35^\circ$ $= 2.664 \times 10^{-22} \text{ kg m s}^{-1}$ momentum of γ ray photon = $\sqrt{4.102^2 + 2.664^2} \times 10^{-22} = 4.891 \times 10^{-22} \text{ kg m s}^{-1}$ wavelength of γ ray photon = $\frac{h}{p} = \frac{6.63 \times 10^{-34}}{4.891 \times 10^{-22}}$ $= 1.36 \times 10^{-12} \text{ m}$ <i>There were good responses but many were unsure of the relationships regarding momentum.</i>	
	(e)	At time $t = 0$ s, a sample consists only of the isotope $^{214}_{82}\text{Pb}$.		

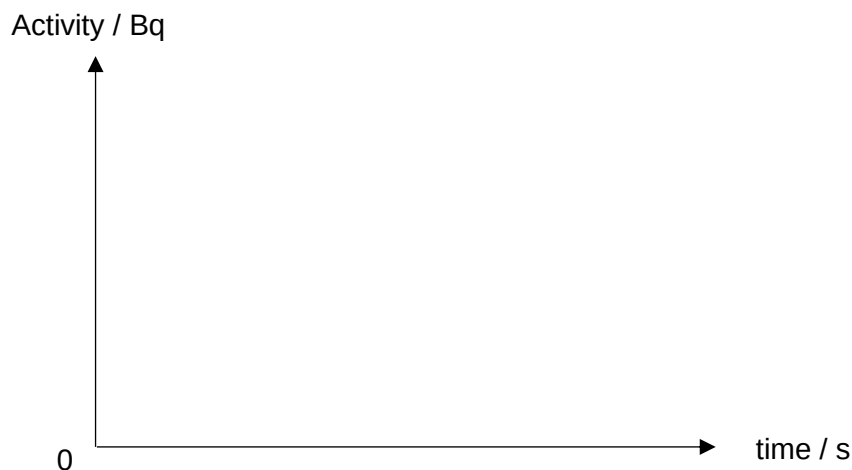


Fig. 9.2

- (i) Without numerical values, sketch on Fig. 9.1 a graph of the activity of $^{214}_{82}\text{Pb}$ with time. Label this graph Pb. [1]
- (ii) Without numerical values, sketch on Fig. 9.1 a graph of the activity of $^{214}_{83}\text{Bi}$ with time. Label this graph Bi. [2]

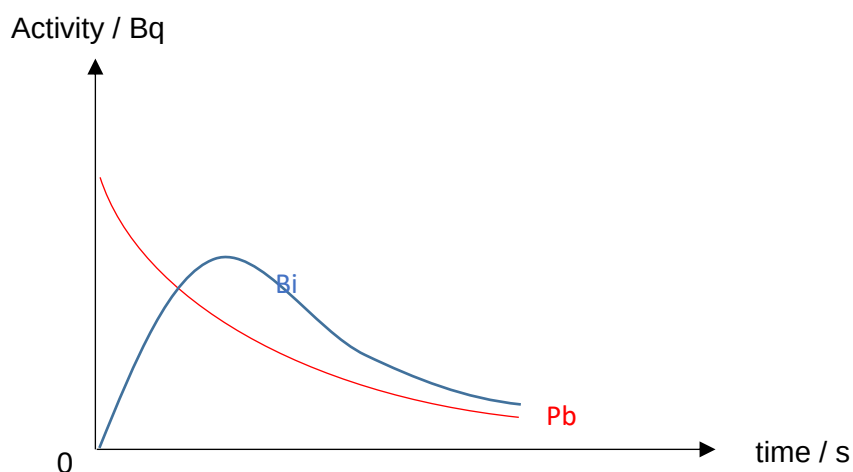


Fig. 9.2

Bi : Starts from zero, Turning point

Most were able to produce the graph for Pb, but responses for Bi often neglected to include the turning point. Many thought the amount of Bi would increase continuously without considering that Bi would undergo decay.

- (f) A sample of $^{214}_{82}\text{Pb}$ has mass $3.5 \mu\text{g}$ at time $t = 0 \text{ s}$. The half-life of $^{214}_{82}\text{Pb}$ is 27 minutes.

- (i) Show that the sample contains approximately 9.8×10^{15} atoms.

1 mole of $^{214}_{82}\text{Pb}$ is 214 g.

			Hence number of moles = $\frac{3.5 \times 10^{-6}}{214}$ Number of atoms = $\frac{3.5 \times 10^{-6}}{214} \times 6.02 \times 10^{23}$ $= 9.8 \times 10^{15}$ <i>Generally well done.</i>	
		(ii)	Show that its decay constant is $4.3 \times 10^{-4} \text{ s}^{-1}$.	
				[1]
			$\lambda = \frac{\ln 2}{27 \times 60}$ $= 4.3 \times 10^{-4} \text{ s}^{-1}$ <i>Generally well done.</i>	
		(iii)	Calculate its activity at time $t = 0 \text{ s}$.	
			activity = Bq	[1]
			$A_0 = \lambda N = 4.3 \times 10^{-4} \times 9.8 \times 10^{15}$ $= 4.2 \times 10^{12} \text{ Bq}$ <i>Generally well done.</i>	
		(iv)	Hence, calculate the time at which its activity has fallen to $8.8 \times 10^9 \text{ Bq}$.	
			time = min	[2]
			$A = A_0 e^{-\lambda t}$ $8.8 \times 10^9 = 4.2 \times 10^{12} e^{-\frac{\ln 2}{27 \times 60} t}$ $t = 14400 \text{ s} = 240 \text{ min}$ <i>Generally well done.</i>	
				[Total: 20]